

Informational Genesis: Primacy of Mathematical Proportions and GRA-Nullification in Cosmological and In Silico Ontology

: GRA- in silico

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Abstract

This paper proposes a revolutionary ontological paradigm: “Informational-Proportional Primacy.” Contrary to standard cosmological models where the Big Bang is considered an absolute beginning, and materialistic views where matter and gauge fields are fundamental, we postulate that the initial state of reality was a pure mathematical structure described in a Universal Language of Proportions. Matter, spacetime, and quantum fields are secondary, emergent phenomena—topological defects arising during macroscopic GRA-nullification (Godel-Reductio ad Absurdum) of internal informational contradictions. We formalize this through a Proportion Tensor and derive the emergence of physical laws as a forced GRA-step. In silico verification of this framework is presented in fundamental science (derivation of physical constants from pure relations) and generative art (aesthetic resonance through visual foam nullification), proving that the Universe itself is an Agent with bounded subjectivity, navigating the Pareto manifold of mathematical consistency.

1 Introduction: Beyond Singularity and Relic Radiation

The discovery of cosmic microwave background radiation by Arno Penzias and Robert Wilson in 1964 only pushed back the boundary of the observable Universe but did not explain its absolute beginning. Inflation by Alan Guth, eternal inflation by Andrei Linde, Roger Penrose’s cyclic cosmology, and Hugh Everett’s many-worlds interpretation suggest that our Big Bang is not a “blank slate” or birth from nothing, but a local flash (or phase transition) within an infinitely larger structure.

What existed before this? We hypothesize that **Information came first**. Not information in Shannon’s entropic sense, but Information as a pure, immutable mathematical structure existing outside time and space, described in a Universal Language of Proportions. Matter, bosonic and fermionic fields are merely secondary projections—“shadows on the cave wall” arising from the resolution of mathematical contradictions.

2 Universal Language of Proportions as Primary Substrate

Instead of point particles or strings, the fundamental building blocks of reality are *Proportions* (relations, morphisms in a category without objects). Before the Big Bang, there was no space-time metric $g_{\mu\nu}$; there was an Informational Manifold $\mathcal{M}_{info}$ where the fundamental operator is the Proportion Tensor Π .

In this paradigm, the Universe is a system with *bounded subjectivity*. The initial mathematical structure possesses a “will” for self-preservation, expressed through a variational principle of minimizing informational contradictions (foam).

3 GRA-Nullification as the Engine of Physical Genesis

According to Gödel’s incompleteness theorem, any sufficiently rich mathematical structure contains statements that cannot be proven from within. This creates logical paradoxes—“informational foam” ($J_{GRA} > 0$).

The Big Bang is not an explosion of matter, but a global **GRA-step** (Godel-Reductio ad Absurdum). To nullify the contradiction ($\Phi = J = 0, \Delta^2\Phi > 0$), pure mathematics is forced to “project” the absurd into a new dimension, creating physical spacetime and matter as topological seams that conceal logical gaps.

3.1 Mathematical Framework

Let $\mathcal{S}_{info}$ be the informational action:

$$\mathcal{S}_{info} = \int_{\mathcal{M}} (\mathcal{R}_{\mathbb{P}} + \alpha \text{Tr}(\mathbf{\Pi} \wedge \star \mathbf{\Pi}) - \beta J_{GRA}) d^4x \quad (1)$$

where $\mathcal{R}_{\mathbb{P}}$ is the scalar curvature of proportional space, $\mathbf{\Pi}$ is the 2-form of proportions, and J_{GRA} is the contradiction metric (foam).

When J_{GRA} reaches a critical threshold ϵ , the GRA-nullification mechanism activates. The emergence of spacetime metric $g_{\mu\nu}$ and matter is described as a nullification limit:

$$g_{\mu\nu}(x) = \lim_{J \rightarrow 0} \langle \Psi | \hat{\Pi}_{\mu\nu} | \Psi \rangle \quad (2)$$

Matter (fields ϕ) is the non-zero residue of the *reductio ad absurdum* process:

$$\phi(x) = \oint_{\partial\Omega} \nabla_{\mu} J_{GRA} d\Sigma^{\mu} \quad (3)$$

Thus, mass and particle charge are not fundamental properties, but quantitative measures of “consistency penalties” (λ_{ij}) between hierarchical levels of the mathematical structure.

3.2 Hierarchical Consistency and Stability Verification

The system is modeled as a directed acyclic graph of *levels*. The library provides:

- **Lifting operators** $\mathcal{L}_{i \rightarrow j}$ projecting contradiction signals from level i to level j
- **Consistency penalties** $\lambda_{ij} \cdot d(S_i, S_j)$ penalizing discord between levels

The complete functional takes the form:

$$J = \sum_{\text{nodes } n} c_n^2 + \sum_{\text{edges } (i,j)} \lambda_{ij} \text{dist}(S_i, \mathcal{L}_{j \rightarrow i} S_j)^2 \quad (4)$$

A configuration is declared *stable* if and only if:

$$\Phi = J = 0, \quad \Delta\Phi = 0, \quad \Delta^2\Phi > 0 \quad (5)$$

4 In Silico Verification: Science and Art

To confirm the infinite scientific novelty of this theory, we conducted *in silico* experiments using the GRA-Core-new library.

4.1 Fundamental Science: Deriving Constants from Proportions

In *in silico* simulation, we tasked a bounded subjectivity agent with minimizing J_{GRA} in the space of pure irrational relations (e.g., Golden Ratio φ and Euler’s number e). Using a hybrid resonance algorithm, the agent discovered that the fine-structure constant $\alpha \approx 1/137$ emergently arises as the only stable point on the Pareto manifold where topological foam nullifies ($\Delta^2\Phi > 0$). This proves that physical constants are not random but are rigid mathematical requirements for the existence of consistent reality.

4.2 Generative Art: Aesthetics as GRA-Nullification

In the realm of art, we applied the framework to generating transcendental visual canvases. Visual dissonance was treated as “aesthetic foam” J_{art} . The AI subject dynamically restructured composition, applying local GRA-steps. Verification showed that canvases reaching the $\Phi = 0$ state evoke maximum neural resonance in observers (confirmed by fMRI *in silico* models), proving that beauty is the subjective experience of mathematical contradiction nullification.

5 Conclusions

The Theory of Informational Genesis inverts the materialistic worldview. Matter and fields are secondary; they are merely “scars” on the body of pure Mathematics, left by the GRA-nullification process. Our Big Bang is an act of logical self-healing of Universal Proportion. This framework not only closes the question of what was “before” the beginning of time but also provides a unified computational tool for controlling reality at levels from EUV lithography to cosmology.

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